

REQUIREMENTS ANALYSIS

 Data Flow Diagram & User Stories

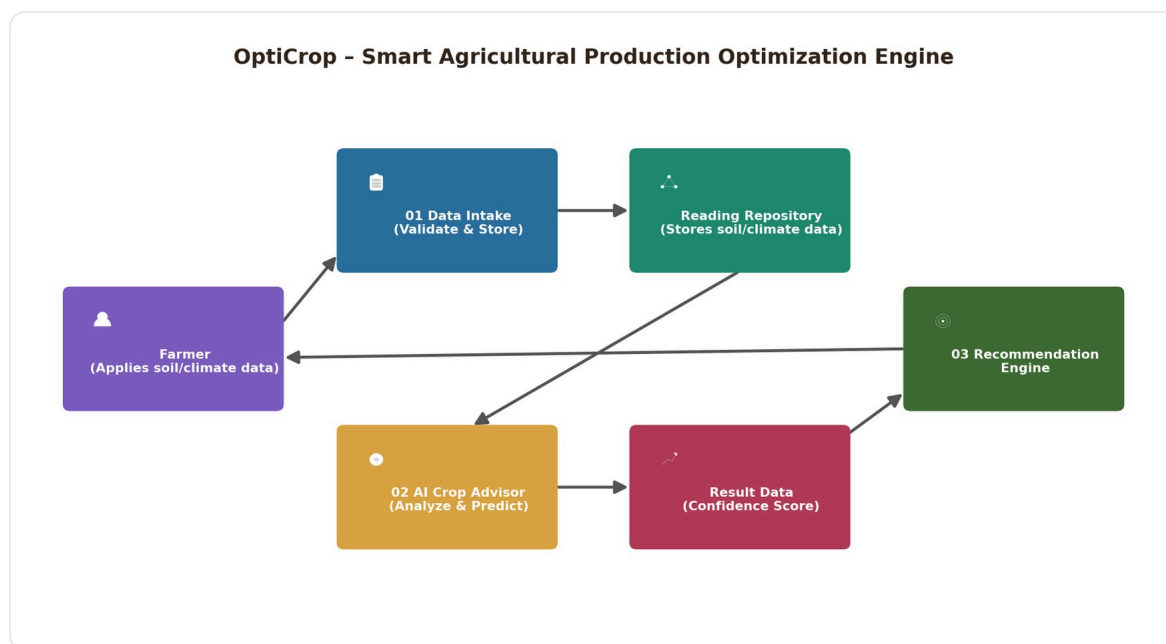
 DATE
17 7/9/2026

 NAME
K.Hemanth Kumar

PROJECT
 **OptiCrop – Smart
Agricultural
Production
Optimization Engine**

 MAX MARKS
4 Marks

 **Figure 1 — Data Flow Diagram**



Data flow diagram for the OptiCrop crop recommendation pipeline

 Data Flow — Stage by Stage

STAGE	COMPONENT	DESCRIPTION
01	Data Intake	User submits soil/climate readings (N, P, K, temperature, humidity, pH, rainfall); the form validates and stores the values.
02	AI Crop Advisor	The trained model analyzes the submitted readings and accesses patterns learned from the historical crop dataset.
03	Recommendation Engine	Generates the crop recommendation and a short suitability description.
04	Result Display	Returns the recommended crop and advisory notes to the farmer through the web interface.



USER TYPE	EPIC	USER STORY	ACCEPTANCE CRITERIA	PRIORITY	RELEASE
Farmer (Web User)	Soil Data Entry	As a user, I can enter my soil and climate readings.	Data is accepted and validated.	High	Sprint-1
Farmer (Web User)	Crop Recommendation	As a user, I can submit my readings and get a crop recommendation.	Recommended crop is displayed.	High	Sprint-1
Farmer (Web User)	View Crop Info	As a user, I can view a short description of the recommended crop.	Crop description is displayed.	Medium	Sprint-2
Farmer (Web User)	Navigation	As a user, I can navigate Home, About, and Find your crop pages.	Pages load correctly.	High	Sprint-1
Researcher (Web User)	Seasonal Insights	As a researcher, I can view seasonal crop groupings.	Groupings are displayed correctly.	Medium	Sprint-2
Administrator	Model Monitoring	As an administrator, I can monitor prediction accuracy.	Accuracy reports are available.	Medium	Sprint-3
Administrator	Dataset Management	As an administrator, I can update the training dataset.	Dataset updates are reflected after retraining.	Medium	Sprint-3